FFSD PROJECT

Domain: Digital Marketplace for Services (non-retail).

Problem Statement: Service Discovery and Booking Platform.

Team Name: 45_ServeEase

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Software Requirements Specification (SRS)

Service Discovery and Booking Platform

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1. Project Overview

In today’s rapidly evolving digital ecosystem, software platforms play a crucial role in simplifying access to essential non-retail services such as plumbing, electrical work, home cleaning, tutoring, and appliance repairs. Customers increasingly rely on online systems to find reliable service providers, yet traditional service discovery methods—manual searches, phone calls, and unverified listings—often lead to inefficiencies, lack of transparency, inconsistent service quality, poor coordination, and an overall negative user experience.

ServeEase – a Digital Marketplace for Services models a real-world service discovery and booking problem using core software engineering principles, focusing on problem domain analysis, stakeholder identification, and clear requirement definition before system design. The system follows a disciplined engineering approach to streamline service discovery, improve booking efficiency, and address real-world service management challenges effectively.

2. Problem Statement Overview

In today's digitally driven economy, customers struggle to access reliable home services such as plumbing, repairs, and tutoring due to fragmented discovery methods. Informal recommendations and scattered listings provide little transparency regarding pricing, service quality, and provider credibility. This lack of clarity leads to scheduling issues for users and operational inefficiencies for service providers, ultimately resulting in dissatisfaction for both parties.

ServeEase – a Service Discovery and Booking Platform addresses these challenges through a centralized digital marketplace for non-retail services. The platform enables users to find verified providers by category and location, view detailed profiles, and manage bookings confidently. With structured tracking and simulated payment handling, ServeEase delivers a scalable, secure, and well-engineered solution that replaces manual processes, improves coordination, and enhances the overall experience for all stakeholders.

3. User Requirements

User requirements define what the system should do from the user's perspective.

Customer Requirements

- The customer shall be able to register, log in, search services by category and location, view provider details, and book services with date and time selection.

- The customer shall be able to track booking history and status, receive notifications, make simulated online payments, and raise support requests.

Service Provider Requirements

- The service provider shall be able to register, manage their profile, services, pricing, availability, and service locations.

- The service provider shall be able to view, accept or reject booking requests, update service completion status, and receive booking and payment notifications.

Customer Support Requirements

- Customer support staff shall be able to view, manage, and resolve booking- and payment-related support tickets.

- Customer support staff shall be able to communicate with customers and service providers and escalate unresolved issues to administrators.

Administrator Requirements

- The administrator shall be able to securely manage customer and service provider accounts, including approval, blocking, and removal.

- The administrator shall be able to manage service categories and monitor system activity through reports.

External System Requirements (Payment Gateway)

- The system shall integrate with a secure external payment gateway to process and validate simulated online payments.

- The system shall record transaction details and notify customers and service providers of payment status.

4. System Requirements

Functional Requirements

1. User Management

- The system shall allow customers to register and log in using valid credentials.
- The system shall allow service providers, customer support staff, and administrators to log in with role-based access.
- The system shall authenticate users using email and password combination
- The system shall restrict access to administrative functions to Admin role only

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2. Browse Services

- The system shall allow customers to browse available services based on category.
- The system shall allow customers to browse services based on location.
- The system shall display service details such as service name, description, pricing, availability, and ratings.
- The system shall allow customers to filter search results by price range (min-max),rating(1-5 stars)

- The system shall allow customers to sort search results by price (ascending/descending), rating (high to low), or distance (nearest first)
- The system shall display for each service: service name, description, base price, provider name, average rating, and distance from search location

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3. Service Provider Discovery

● The system shall allow customers to view a list of service providers for a selected service.

● The system shall display service provider details including provider profile, experience, pricing, and availability.

● The system shall display customer ratings and reviews for service providers.

- The system shall allow service providers to update,deactivate/reactivate service listings

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4. Check Service Availability

● The system shall display provider's available time slots for selected date

●The system shall prevent selection of past dates or time slots

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5. Service Booking

● The system shall allow customers to book services for a selected date and time.
● The system shall generate a service booking request and unique reference number after customer confirmation.
● The system shall notify the selected service provider of new booking requests.

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6. Booking Request Management

- The system shall notify service provider immediately when new booking request is created

● The system shall allow service providers to view incoming booking requests.
● The system shall allow service providers to accept or reject booking requests.
● The system shall update booking status based on the service provider's decision.

- The system shall notify customer when provider accepts or rejects booking request

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7. Booking Modification and Cancellation

● The system shall allow customers to modify or cancel a booking within the allowed policy time.
● The system shall notify service providers of booking modifications or cancellations.

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8. Service Completion Management

● The system shall allow service providers to update service completion status.

● The system shall maintain service records after completion.

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9. Notification Management

● The system shall send notifications to customers and service providers for booking confirmation, acceptance, rejection, and completion.
● The system shall notify users about payment status and support ticket updates.

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10. Payment Processing

● The system shall integrate with a payment gateway for simulated online payments.
● The system shall confirm payment status before finalizing the service booking.
● The system shall generate payment transaction records.

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11. Support Ticket Management

● The system shall allow customers to raise support tickets for service-related or payment-related issues.
● The system shall allow customer support staff to manage and resolve support tickets.

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12. Administrative Management

● The system shall allow administrators to manage customer and service provider accounts.
● The system shall allow administrators to approve, block, or remove service providers.
● The system shall allow administrators to manage service categories.

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13. External Services Integration

● The system shall integrate external notification services for sending alerts and updates.
● The system shall integrate third-party payment systems for transaction processing.

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14. Booking History

- The system shall allow users to view detailed information for any historical booking

Non-Functional Requirements

1. Performance Requirements

● The system shall respond to service search and booking requests within an acceptable time.

● The system shall support multiple concurrent users without performance degradation.
● The system shall handle simultaneous booking requests without data inconsistency.

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2. Availability Requirements

● The system shall be available to users at all times except during scheduled maintenance.
● The system shall ensure uninterrupted access to core functionalities such as service discovery and booking.

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3. Security Requirements

● The system shall enforce secure authentication and authorization mechanisms.
● The system shall protect user credentials and payment-related information.
● Only authorized users shall be allowed to access restricted system features and administrative functions.

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4. Usability Requirements

● The system shall provide a user-friendly and intuitive web interface.
● The system shall be accessible through standard web browsers.
● The system shall ensure that service booking and payment processes can be completed in minimal steps.

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5. Reliability Requirements

● The system shall ensure consistency of booking and payment data.
● The system shall prevent duplicate or invalid service bookings.
● The system shall maintain correct system behavior during concurrent operations.

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6. Scalability Requirements

● The system shall support an increasing number of customers and service providers.
● The system shall allow the addition of new service categories without affecting existing functionality.

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7. Maintainability Requirements

● The system shall be designed using modular architecture to support easy maintenance.
● The system shall allow updates to service rules, pricing models, and policies with minimal system changes.

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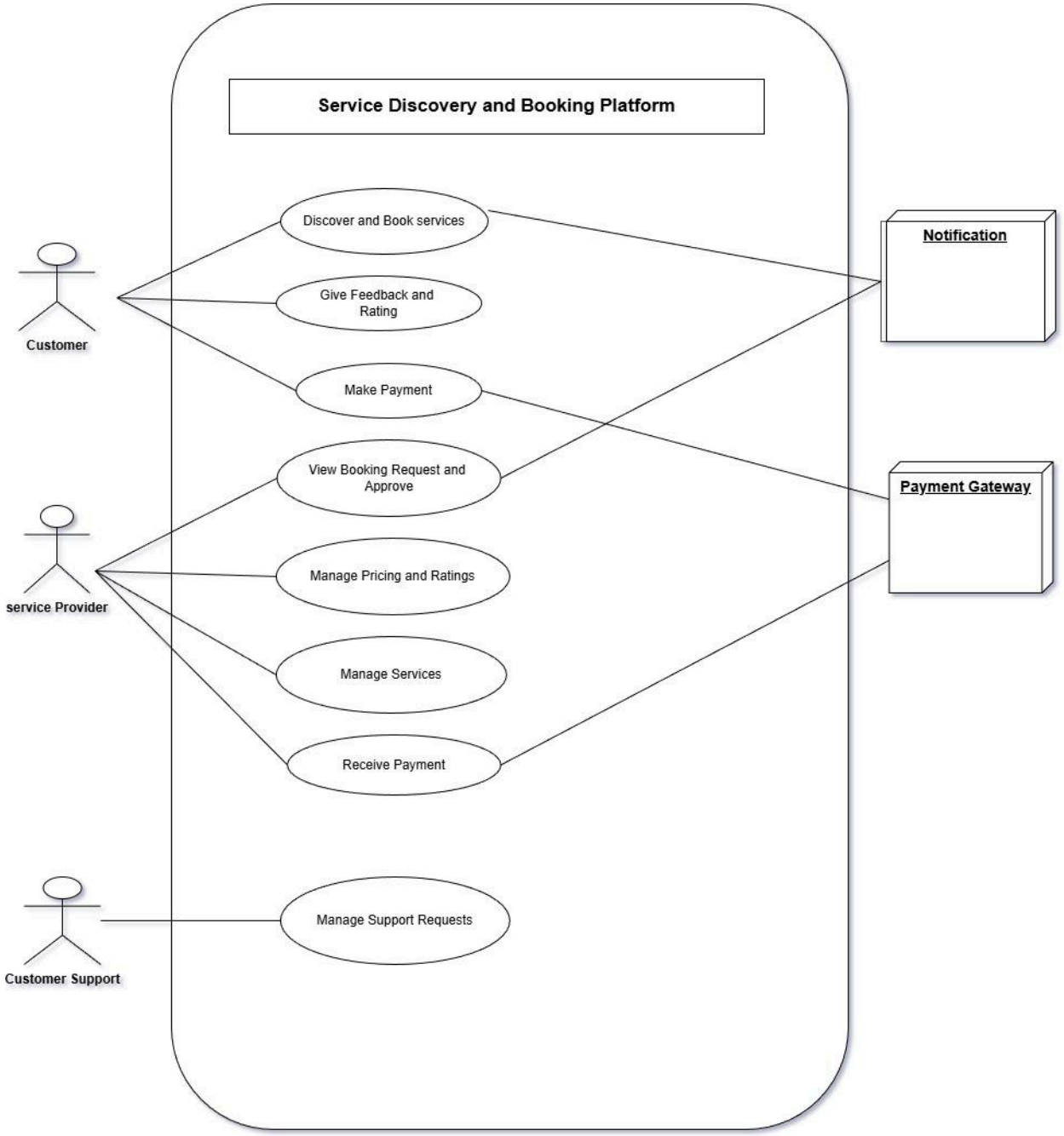
8. Compatibility Requirements

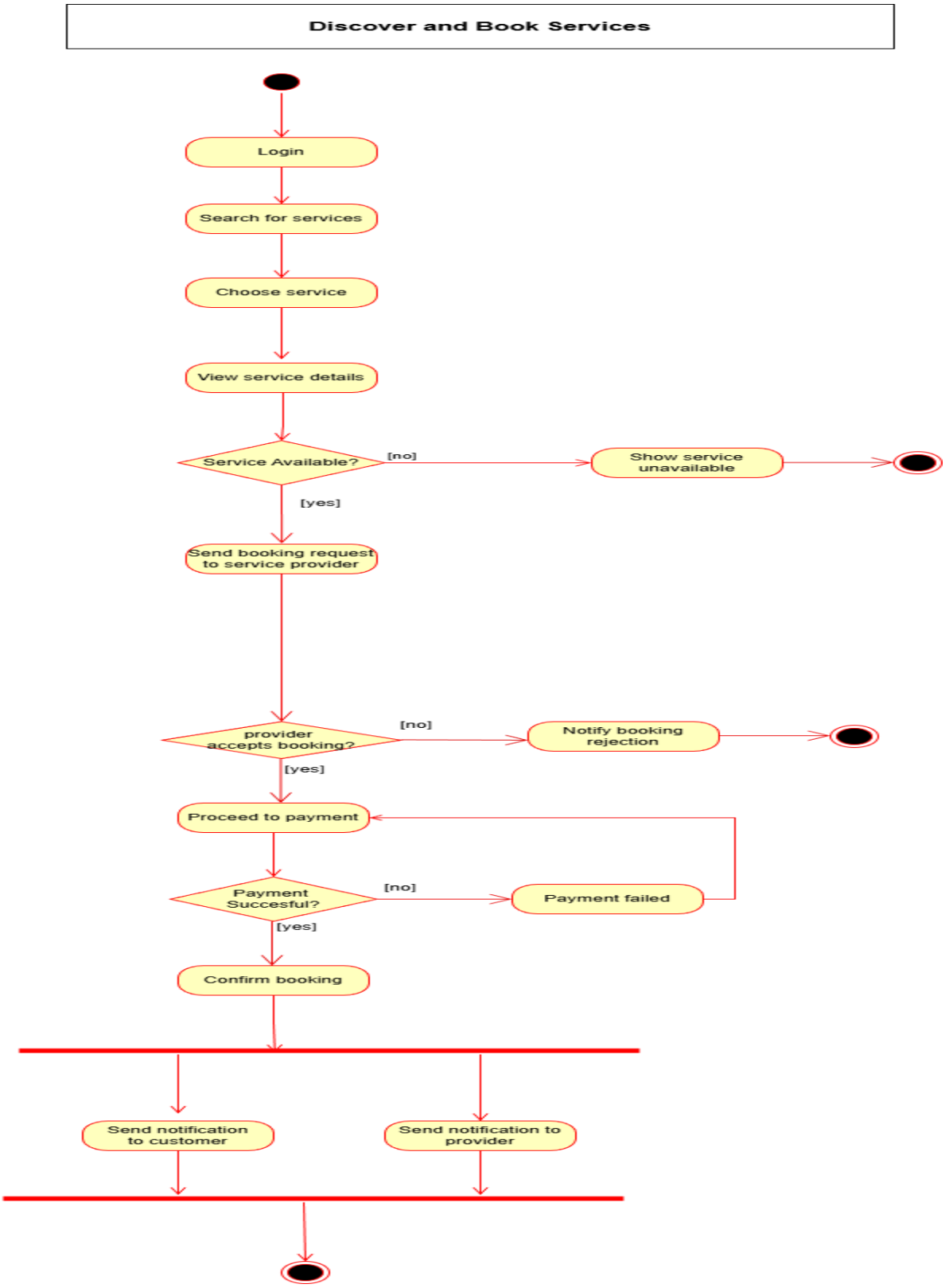
- The system shall be compatible with modern web browsers.
- The system shall integrate with external systems such as payment gateways and notification services.

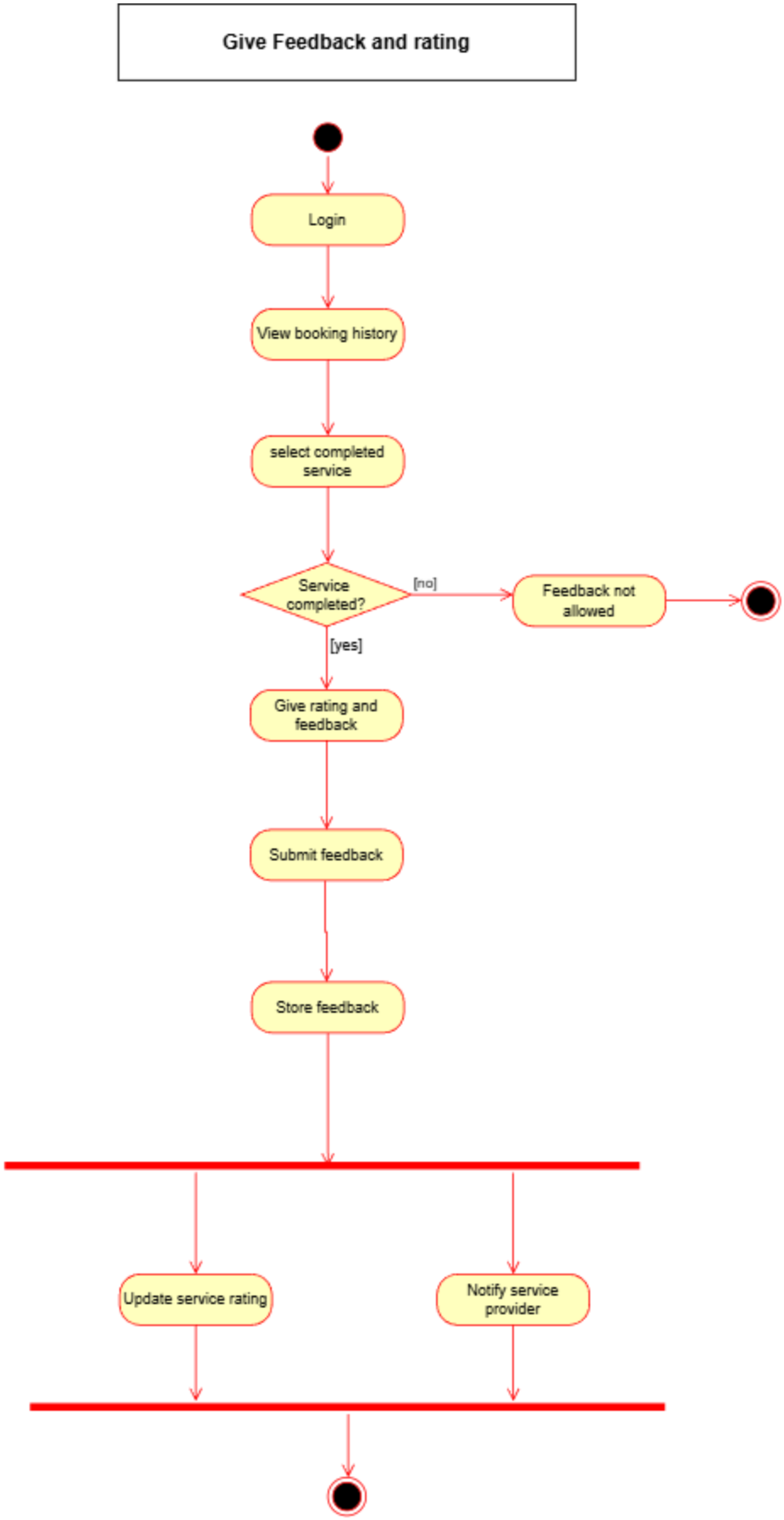
5) UML Diagrams

a) Use-case diagram

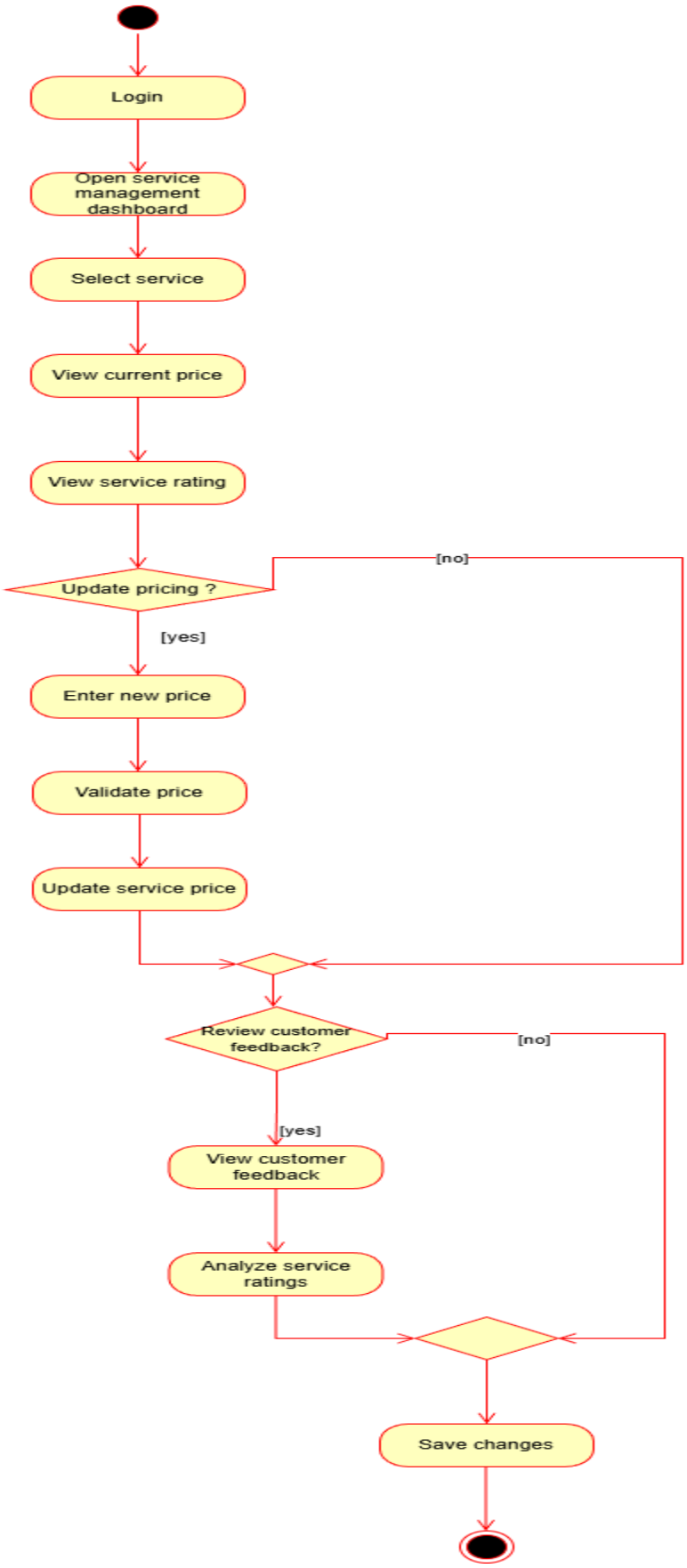


b) Activity Diagrams

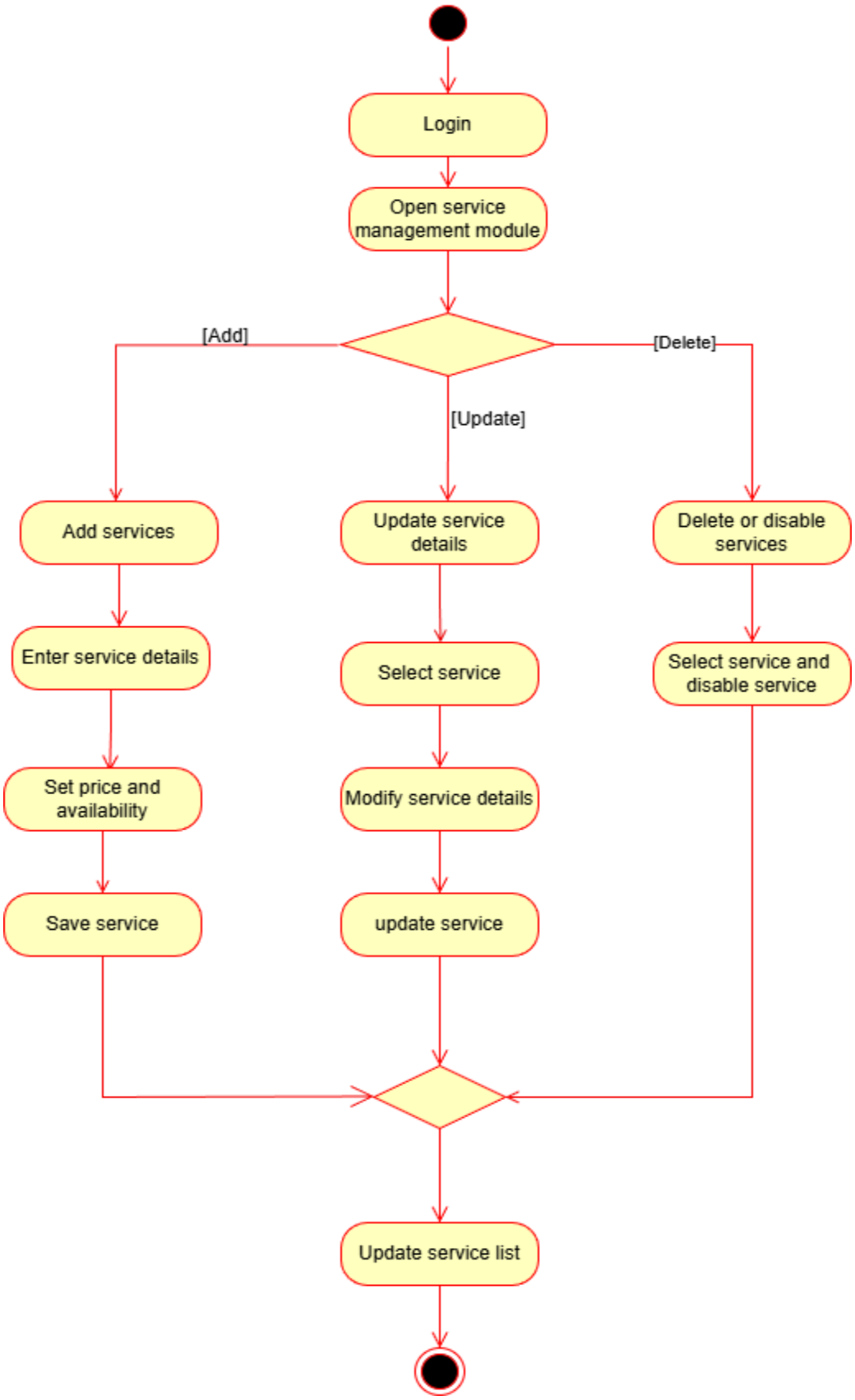


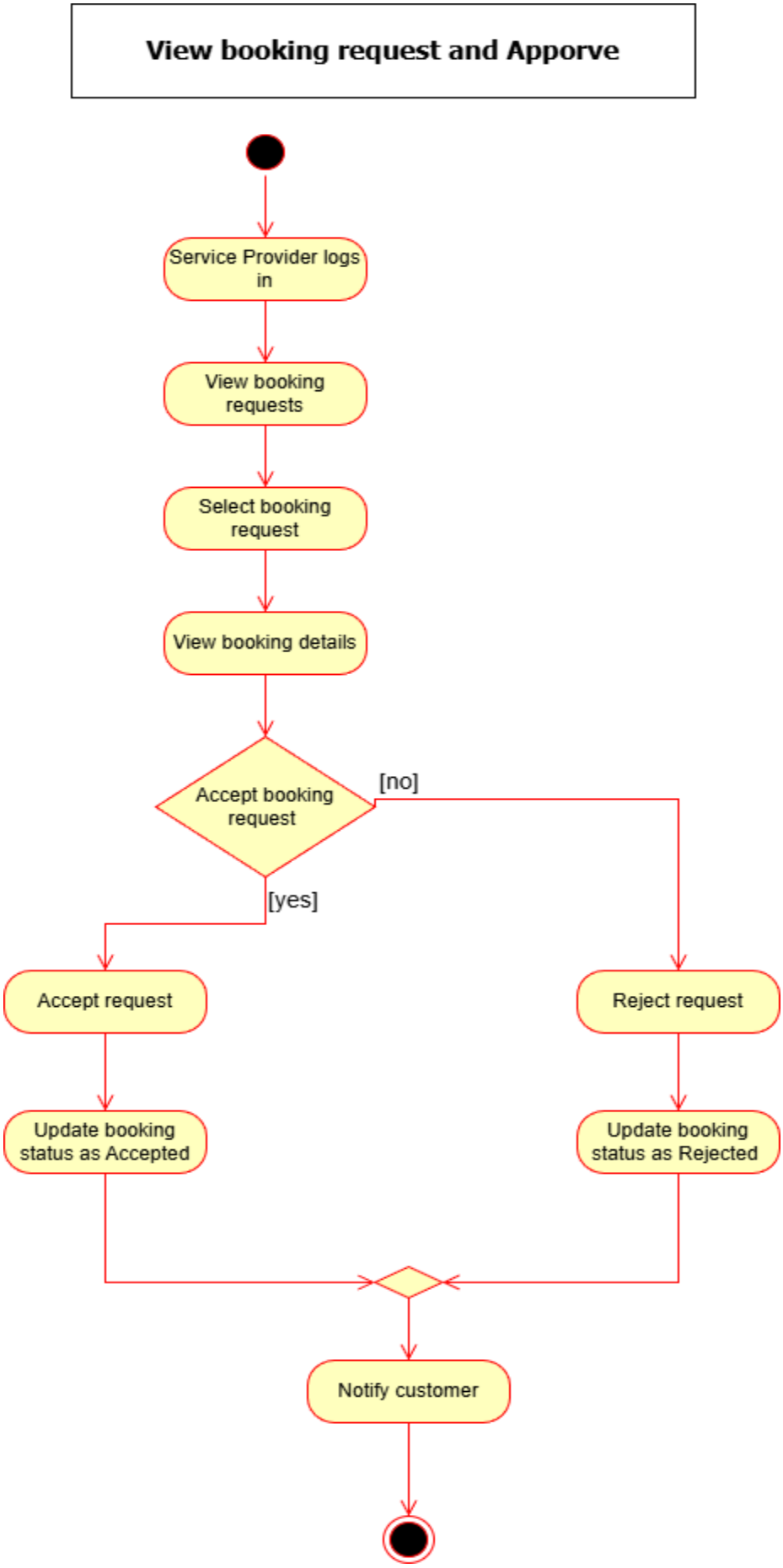

Give Feedback and rating
Login
View booking history
select completed service
Service completed?
[no]
Feedback not allowed
[yes]
Give rating and feedback
Submit feedback
Store feedback
Update service rating
Notify service provider

Manage pricing and ratings

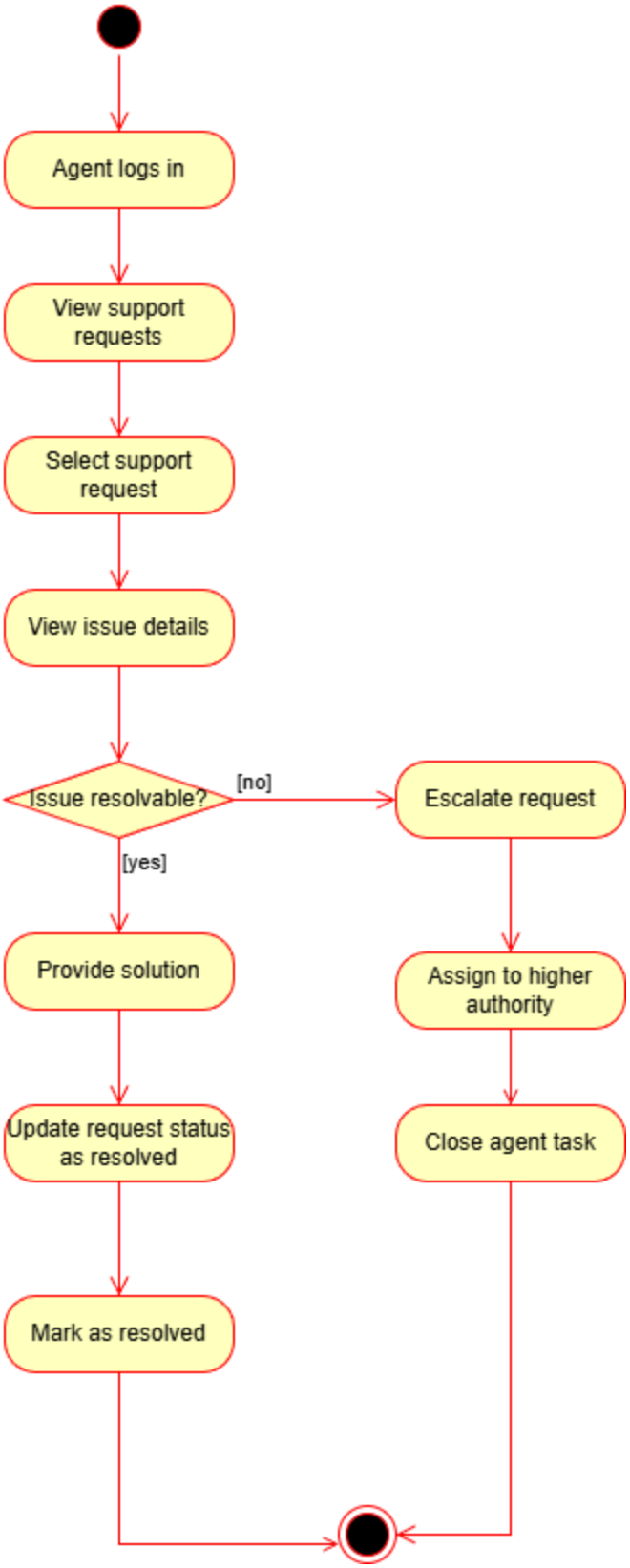


Manage Services

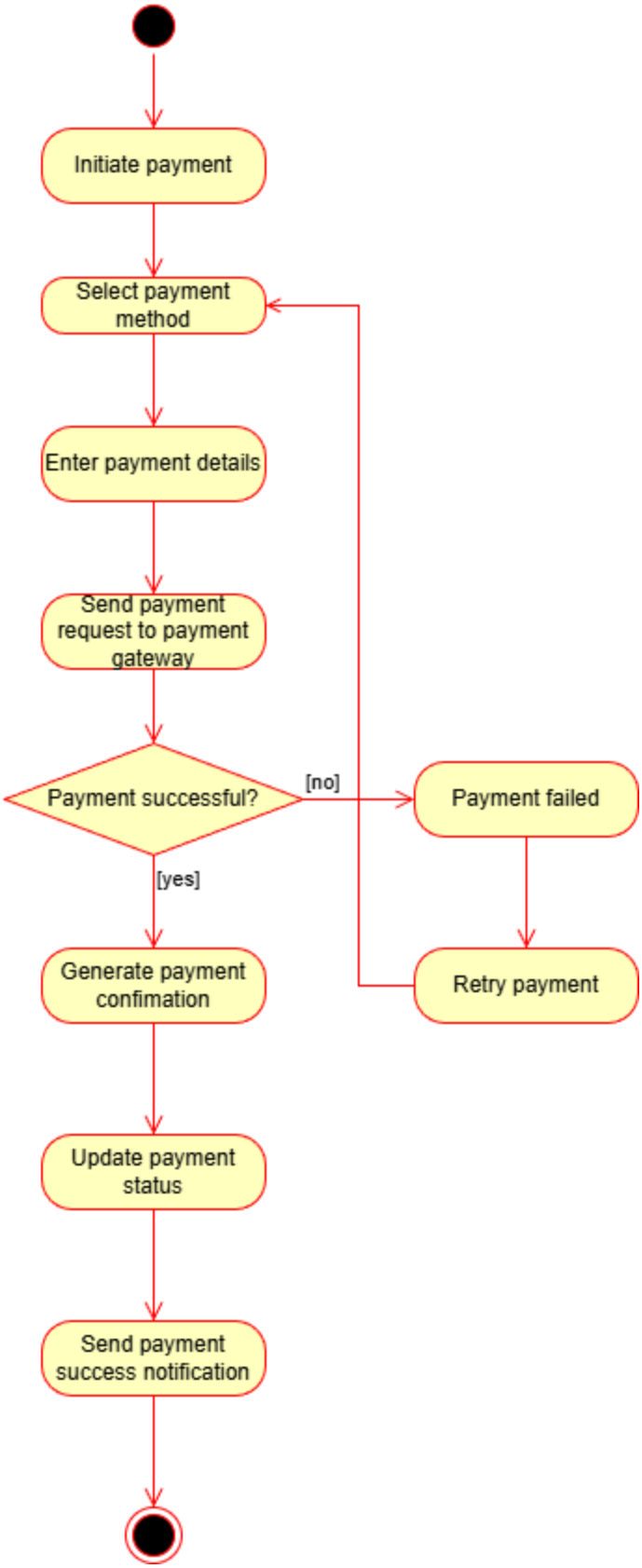



View booking request and Apporve
Service Provider logs in
View booking requests
Select booking request
View booking details
Accept booking request
[no]
[yes]
Accept request
Reject request
Update booking status as Accepted
Update booking status as Rejected
Notify customer

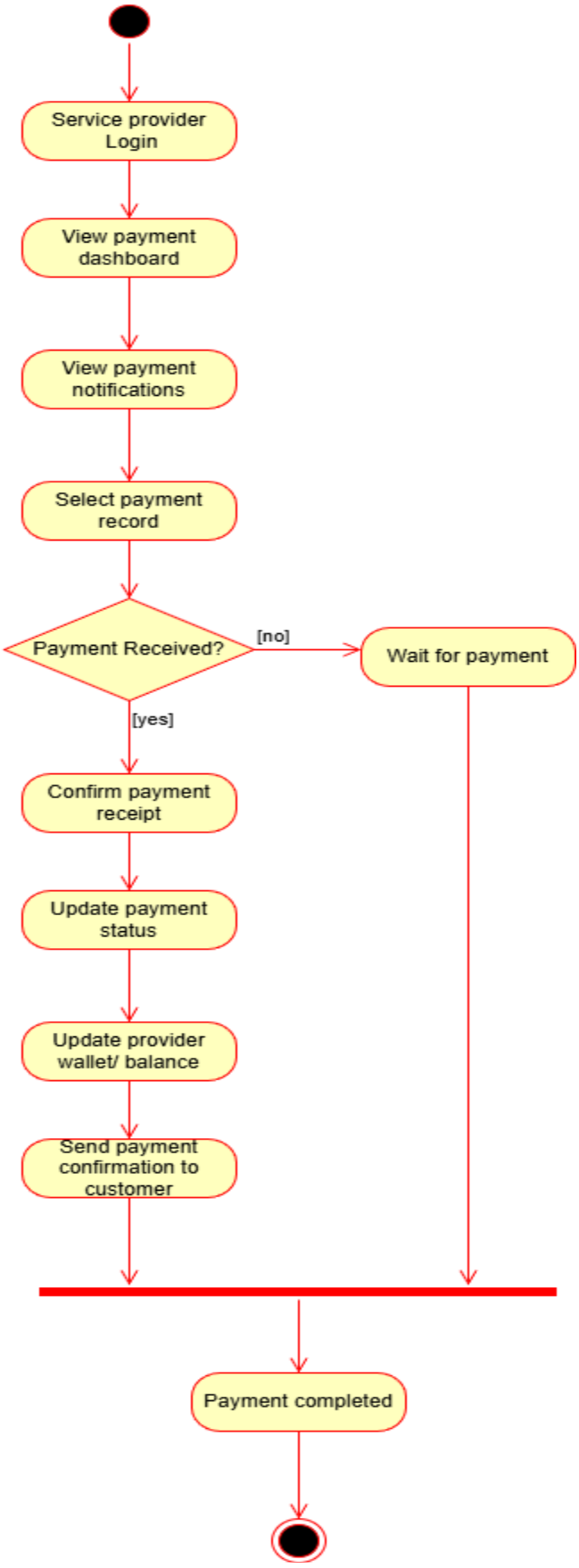
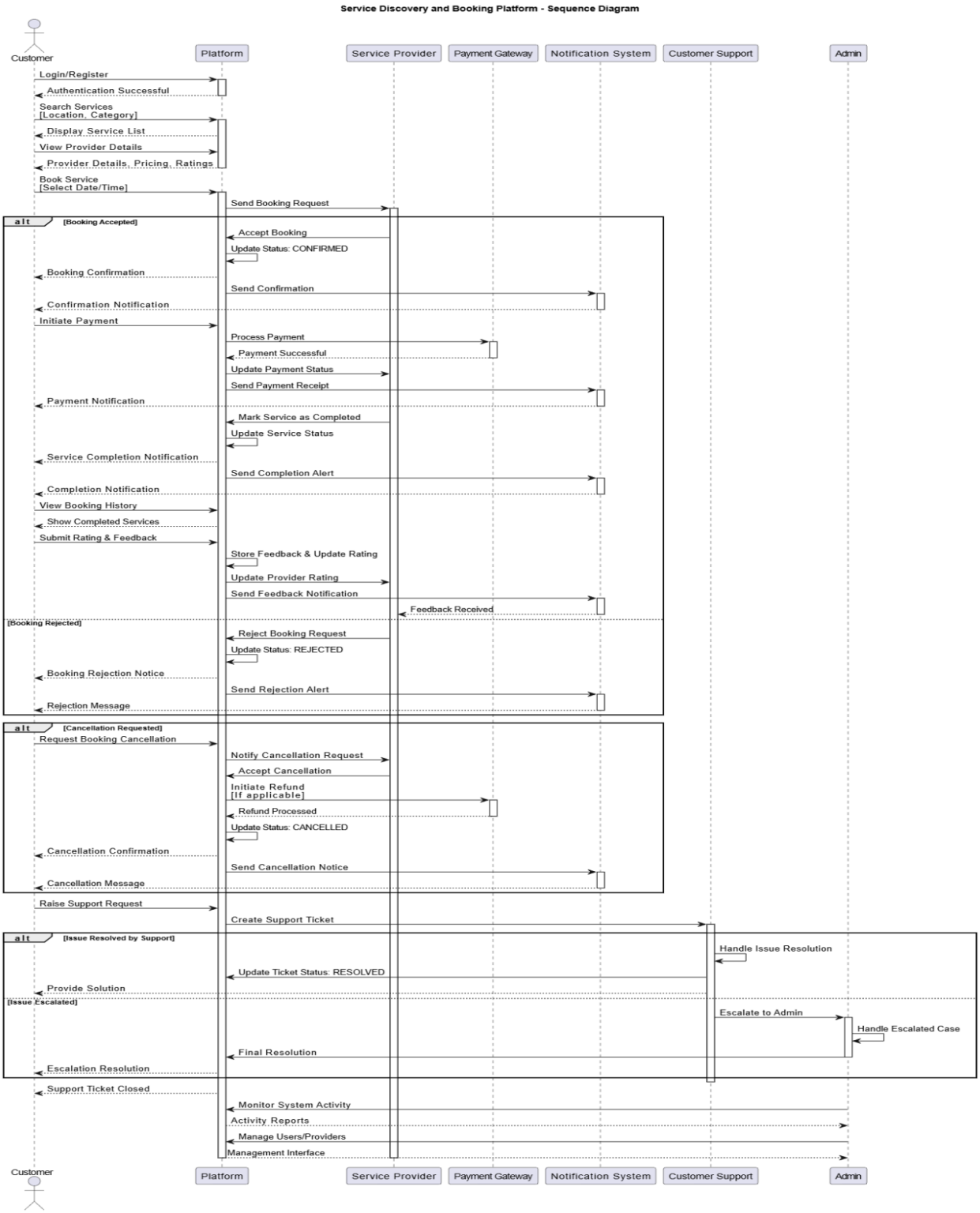
Manage Support Requests



Make Payment



Receive Payment



c) Sequence diagram